



Yakeen Plus (2025)

Practice Test-19

DURATION : 200 Minutes

DATE : 30/12/2024

M. MARKS : 720

Topics Covered

Physics :	Gravitation, Properties of Solids and Liquids, Thermodynamics, Kinetic Theory of Gases Oscillation and Waves
Chemistry :	Redox Reaction, Electrochemistry, Hydrocarbon, Haloalkanes and Haloarenes, Coordination Compound
Biology :	(Botany): Morphology of Flowering Plants, Anatomy of Flowering Plants, Respiration in Plants Photosynthesis in Higher Plants, Plant Growth and Development (Zoology): Body Fluids and Circulation, Excretory Products & their Elimination, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours 20 min.** duration.
3. The test booklet consists of **200** questions. The maximum marks are **720**.
4. There are **four Section** in the Question Paper, **Section I, II, III & IV** consisting of Section-I (Physics from question number **1 to 50**), Section-II (Chemistry: from question number **51 to 100**), Section-III (Botany: from question number **101 to 150**) & Section IV (Zoology: from question number **151 to 200**), having 50 Questions in each Subject and each subject is divided in two Section, Section A consisting 35 questions (all questions are compulsory) and Section B consisting 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

Practice Test-19

Yakeen Plus (2025)

Q1

SECTION - A (PHYSICS)

Following statements are given:

- A. The average kinetic energy of a gas molecule decreases when the temperature is reduced.
- B. The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.
- C. The average kinetic energy of a gas molecule decreases with increase in volume.
- D. The volume of gas decreases with increase in temperature.

Choose the **correct** answer from the options given below:

- (1) A only
- (2) A, B and D only
- (3) B and D only
- (4) A, B and C only

Q2 Given below are two statements :

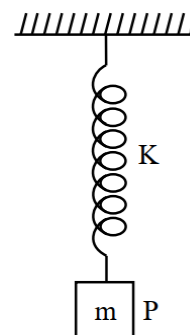
Statement I: The contact angle between a solid and a liquid is a property of the material of the solid and liquid as well.

Statement II: The rise of a liquid in a capillary tube does not depend on the inner radius of the tube.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q3 In simple harmonic motion, the total mechanical energy of given system is E . If mass of oscillating particle P is doubled then the new energy of the system for same amplitude is:



- (1) $E\sqrt{2}$
- (2) E
- (3) $2E$
- (4) $E/\sqrt{2}$

Q4 When a solid ball of volume V is falling through a viscous liquid, a viscous force F acts on it. If another ball of volume $2V$ of the same material is falling through the same liquid then the viscous force experienced by it will be: (both fall with terminal velocities)

- (1) F
- (2) $\frac{F}{2}$
- (3) $2F$
- (4) $\frac{F}{4}$

Q5 The translational degrees of freedom (f_t) and rotational degrees of freedom (f_r) of CH_4 molecule are:

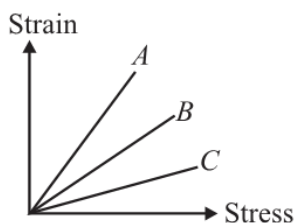
- (1) $f_t = 3$ and $f_r = 3$
- (2) $f_t = 2$ and $f_r = 3$
- (3) $f_t = 2$ and $f_r = 2$
- (4) $f_t = 3$ and $f_r = 2$

Q6 The length of a second pendulum at a height $h = 2R$ from earth surface will be : (Given R = Radius of earth and acceleration due to gravity at the surface of earth, $g = \pi^2 \text{ ms}^{-2}$)

- (1) $\frac{2}{9}m$
- (2) $\frac{4}{9}m$
- (3) $\frac{8}{9}m$
- (4) $\frac{1}{9}m$

Q7 The strain-stress curves of three wires of different material is shown in the figure:

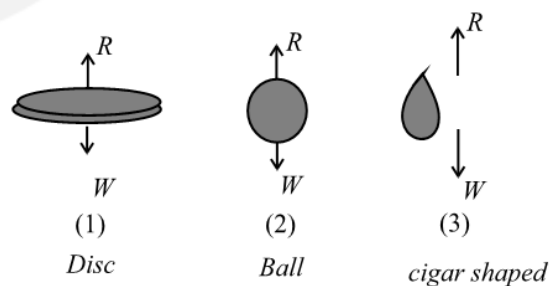




- (1) Young's modulus of wire A is maximum.
 (2) Young's modulus of wire B is maximum.
 (3) Young's modulus of wire C is maximum.
 (4) Young's modulus of all wires are same.
- Q8** An object is taken to a height above the surface of earth at a distance $\frac{5}{4}R$ from the centre of the earth. Where radius of earth, $R = 6400$ km. The percentage decrease in the weight of the object will be:
 (1) 36% (2) 50%
 (3) 64% (4) 25%
- Q9** Two soap bubbles of radii R_1 and R_2 are kept in vacuum at constant temperature, the ratio of masses of air inside then is:
 (1) $\frac{R_1}{R_2}$ (2) $\frac{R_1^3}{R_2^3}$
 (3) $\frac{R_1^2}{R_2^2}$ (4) $\frac{R_2^3}{R_1^3}$
- Q10** A balloon filled with helium (32°C and 1.7 atm.) bursts. Immediately afterwards the expansion of helium can be considered as:
 (1) reversible isothermal.
 (2) irreversible isothermal.
 (3) reversible adiabatic.
 (4) irreversible adiabatic.
- Q11** When a particle executes Simple Harmonic Motion, the nature of graph of velocity as a function of displacement will be?
 (1) Circular
 (2) Elliptical
 (3) Sinusoidal
 (4) Straight line
- Q12** 7 mol of a certain monatomic ideal gas undergoes a temperature increase of 40K at constant volume. The increase in the internal energy of the gas in this process is:
 (Given $R = 8.3\text{ JK}^{-1}\text{ mol}^{-1}$)
 (1) 5810 J (2) 3486 J

(3) 11620 J (4) 6972 J

- Q13** A monatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be:
 (1) P (2) $8P$
 (3) $32P$ (4) $64P$
- Q14** Water is flowing in a pipe of diameter 4 cm with a velocity 3 m/s. The water then enters into a tube of diameter 2 cm. The velocity of water in the other pipe is:
 (1) 3 m/s
 (2) 6 m/s
 (3) 12 m/s
 (4) 8 m/s
- Q15** The percentage decrease in the weight of a rocket, when taken to a height of 32 km above the surface of earth will be:
 (Radius of earth = 6400 km)
 (1) 1% (2) 3%
 (3) 4% (4) 0.5%
- Q16** When a body falls in air, the resistance of air depends to a great extent on the shape of the body, 3 different shapes are given. Identify the combination of air resistances which truly represents the physical situation. (The cross-sectional areas are the same).



- (1) $1 < 2 < 3$
 (2) $2 < 3 < 1$
 (3) $3 < 2 < 1$
 (4) $3 < 1 < 2$

- Q17** A gas has n degree of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be:
 (1) $\frac{n}{n+2}$ (2) $\frac{n+2}{n}$



(3) $\frac{n}{2n+2}$

(4) $\frac{n}{n-2}$

Q18 A transverse wave is represented by $y = 2\sin(\omega t - kx)$ cm. The value of wavelength (in cm) for which the wave velocity becomes equal to the maximum particle velocity, will be:

- (1) 4π (2) 2π
(3) π (4) 2

Q19 If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately:

- (1) decrease by 2% (2) decrease by 4%
(3) increase by 2% (4) increase by 4%

Q20 The force required to stretch a wire of cross-section 1 cm^2 to double its length will be:

(Given Young's modulus of the wire = $2 \times 10^{11} \text{ N/m}^2$)

- (1) $1 \times 10^7 \text{ N}$
(2) $1.5 \times 10^7 \text{ N}$
(3) $2 \times 10^7 \text{ N}$
(4) $2.5 \times 10^7 \text{ N}$

Q21 Given below are two statements:

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is ν . If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become 2ν .

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q22 In the wave equation

$$y = 0.5 \sin \frac{2\pi}{\lambda} (400t - x) \text{ m}$$

The velocity of the wave will be:

(1) 200 m/s

(2) $200\sqrt{2} \text{ m/s}$

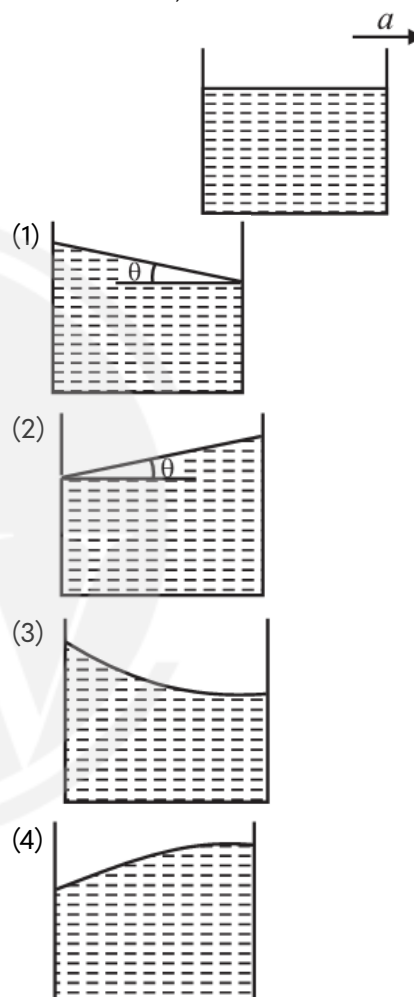
(3) 400 m/s

(4) $400\sqrt{2} \text{ m/s}$

Q23 Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. The ratio of their volumes is:

- (1) 4 : 1 (2) 2 : 1
(3) 0.8 : 1 (4) 8 : 1

Q24 Container filled with water is accelerating with acceleration α , then water surface will look like:



Q25 When the temperature of a metal wire is increased from 0°C to 10°C , its length increases by 0.02%. The percentage change in its mass density will be closest to:

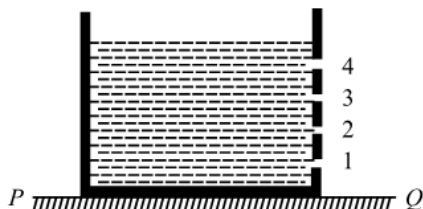
- (1) 0.06 (2) 0.008
(3) 2.3 (4) 0.8

Q26 A sample contains mixture of helium and oxygen gas. The ratio of root mean square speed of helium and oxygen in the sample, is:



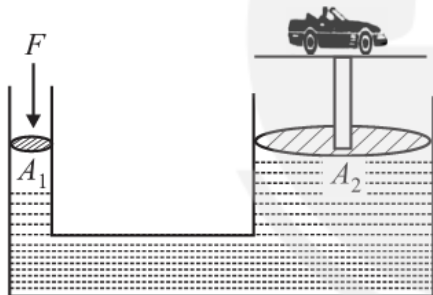
- (1) $\frac{1}{32}$ (2) $\frac{1}{2\sqrt{2}}$
 (3) $\frac{1}{4}$ (4) $\frac{2\sqrt{2}}{1}$

Q27 A cylindrical vessel of 90 cm height is kept filled upto the brim. It has four holes 1, 2, 3, 4 which are respectively at heights of 20 cm, 30 cm, 45 cm and 50 cm from the horizontal floor PQ . The water falling at the maximum horizontal distance from the vessel comes from:



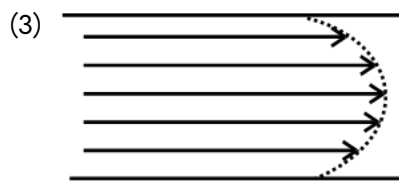
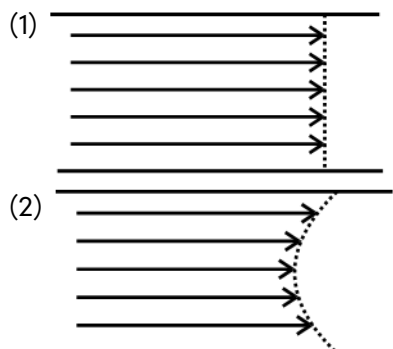
- (1) Hole number 4
 (2) Hole number 3
 (3) Hole number 2
 (4) Hole number 1

Q28 In hydraulic lift area of two cross-section are $A_1 = 1 \text{ cm}^2$ and $A_2 = 1000 \text{ cm}^2$ respectively, then force F required to lift 20,000 kg will be:



- (1) 1 N (2) 2 N
 (3) 4 N (4) 20 N

Q29 A viscous fluid is flowing through a cylindrical tube. The velocity distribution of the fluid is best represented by the diagram.



- (4) None of these

Q30 A body of weight 200 N is suspended from a tree branch through a chain of mass 10 kg. The branch pulls the chain by a force equal to (if $g = 10 \text{ m/s}^2$):

- (1) 300 N (2) 150 N
 (3) 100 N (4) 200 N

Q31 Pressure inside a soap bubble is greater than the pressure outside by an amount:
 (Given : R = Radius of bubble, S = Surface tension of bubble)

- (1) $\frac{4R}{S}$ (2) $\frac{4S}{R}$
 (3) $\frac{2S}{R}$ (4) $\frac{S}{R}$

Q32 The volume of an ideal gas ($\gamma = 1.5$) is changed adiabatically from 5 litres to 4 litres. The ratio of initial pressure to final pressure is:

- (1) $\frac{4}{5}$ (2) $\frac{8}{5\sqrt{5}}$
 (3) $\frac{16}{25}$ (4) $\frac{2}{\sqrt{5}}$

Q33 The excess pressure inside a soap bubble is thrice the excess pressure inside a second soap bubble. The ratio between the volume of the first and the second bubble is:

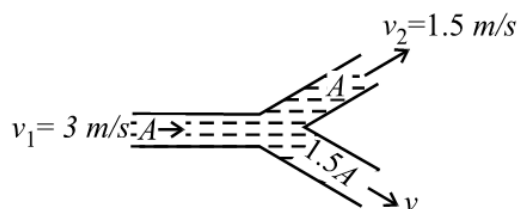
- (1) 1 : 81 (2) 1 : 9
 (3) 1 : 27 (4) 1 : 3

Q34 The work done in blowing a soap bubble of volume ' V ' is W . The work done in blowing a soap bubble of volume ' $2V$ ' is:

- (1) W (2) $2W$
 (3) $2\frac{2}{3}W$ (4) $3\frac{2}{3}W$

Q35 An incompressible liquid flow through a horizontal tube as shown in the following figure. Then the velocity v of the fluid is:



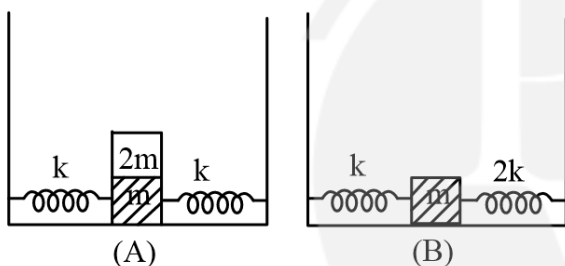


- (1) 3.0 m/s
(2) 1.5 m/s
(3) 1.0 m/s
(4) 2.25 m/s

Q36

SECTION - B (PHYSICS)

In figure (A), mass '2m' is fixed on mass 'm' which is attached to two springs of spring constant k . In figure (B), mass 'm' is attached to two springs of spring constant ' k ' and ' $2k$ '. If mass 'm' in (A) and in (B) are displaced by distance ' x ' horizontally and then released, then time period T_1 and T_2 corresponding to (A) and (B) respectively following the relation.

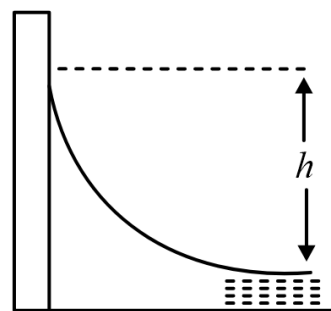


- (1) $\frac{T_1}{T_2} = \frac{3}{\sqrt{2}}$ (2) $\frac{T_1}{T_2} = \sqrt{\frac{3}{2}}$
(3) $\frac{T_1}{T_2} = \sqrt{\frac{2}{3}}$ (4) $\frac{T_1}{T_2} = \frac{\sqrt{2}}{3}$

Q37 The lower end of a capillary tube of radius r is placed vertically in water, then with the rise of water in the capillary heat evolved is: (h is capillary rise and J mechanical equivalent of heat)

- (1) $\frac{\pi r^2 h^2 \rho g}{2J}$
(2) $\frac{\pi r^2 h^2 \rho g}{J}$
(3) $-\frac{\pi r^2 h^2 \rho g}{3J}$
(4) $-\frac{r^2 h^2 \rho g}{J}$

Q38 Water in a clean aquarium forms a meniscus, as shown in figure. The difference in height h between the centre and the edge of the meniscus is: (The surface tension of water is T and density of liquid is ρ)



- (1) $\sqrt{\frac{2T}{\rho g}}$ (2) $\sqrt{\frac{T}{\rho g}}$
(3) $\sqrt{\frac{4T}{\rho g}}$ (4) None of these

Q39 A gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T . Assuming the gases to be ideal and the oxygen bond to be rigid, the total internal energy (in units of RT) of the mixture is:

- (1) 11 (2) 13
(3) 15 (4) 20

Q40 Two identical strings X and Z made of same material have tension T_x and T_z in them. If their fundamental frequencies are 450 Hz and 300 Hz, respectively, then the ratio T_x/T_z is:

- (1) 2.25 (2) 1.25
(3) 0.44 (4) 1.5

Q41 A small ball of mass m and density ρ is dropped in a viscous liquid of density ρ_0 . After sometime, the ball falls with constant velocity. The viscous force on the ball is:

- (1) $mg \left(\frac{\rho_0}{\rho} - 1 \right)$
(2) $mg(1 - \rho\rho_0)$
(3) $mg \left(1 + \frac{\rho}{\rho_0} \right)$
(4) $mg \left(1 - \frac{\rho_0}{\rho} \right)$

Q42 An ideal gas in a closed container is slowly heated. As its temperature increases, which of the following statements are **true**?

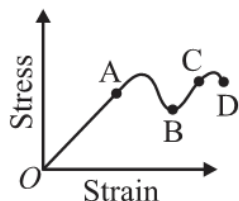
- A. the mean free path of the molecules decreases.
B. the mean collision time between the molecules decreases.
C. the mean free path remains unchanged.
D. the mean collision time remains unchanged.
(1) B and C (2) A and B



(3) C and D

(4) A and D

- Q43** A graph is shown between stress and strain for a metal. The part in which Hooke's law holds good is:



(1) OA

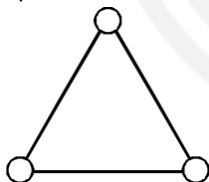
(2) AB

(3) BC

(4) CD

- Q44** A uniform thin rope of length 12 m and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wavetrain of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope ?
- (1) 12 (2) 3
(3) 9 (4) 6

- Q45** Consider a gas of triatomic molecules. The molecules are assumed to be triangular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is:

(1) $\frac{3}{2}RT$ (2) $3RT$ (3) $\frac{5}{2}RT$ (4) $\frac{9}{2}RT$

- Q46** A liquid is filled into a semi elliptical cross-section with a as semi-major axis and b as semi-minor axis. The ratio of surface tension forces on the curved part and plane part of the tube in vertical position will be:
- (1) $\frac{\pi(a+b)}{4b}$ (2) $\frac{2\pi a}{b}$
(3) $\frac{\pi a}{4b}$ (4) $\frac{\pi(a-b)}{4b}$
- Q47** Assume there are two identical simple pendulum clocks. Clock - 1 is placed on the earth and Clock - 2 is placed on a space station located at a

height h above the earth surface. Clock - 1 and Clock - 2 operate at time periods 4 s and 6 s respectively. Then the value of h is: (consider radius of earth $R_E = 6400$ km and g on earth 10 m/s^2)

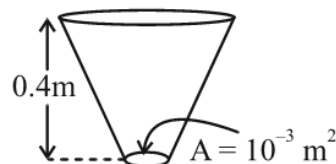
(1) 1200 km

(2) 1600 km

(3) 3200 km

(4) 4800 km

- Q48** A uniformly tapering vessel is filled with a liquid of density 900 kg/m^3 . The force that acts on the base of the vessel due to the liquid is:



(1) 3.6 N

(2) 7.2 N

(3) 9.0 N

(4) 14.4 N

- Q49** For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5m, while the distance between one crest and one trough is 1.5m. The possible wavelengths (in m) of the waves are:
- (1) 1, 3, 5, (2) 1, 2, 3,
(3) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \dots$ (4) $\frac{1}{1}, \frac{1}{3}, \frac{1}{5}, \dots$

- Q50** A spherical ball of radius $1 \times 10^{-4} \text{ m}$ and density 10^5 kg/m^3 falls freely under gravity through a distance h before entering a tank of water. If after entering in water the velocity of the ball does **not** change, then the value of h is approximately: (The coefficient of viscosity of water is $9.8 \times 10^{-6} \text{ N s/m}^2$)
- (1) 2296 m (2) 2518 m
(3) 2249 m (4) 2396 m

Q51**SECTION - A (CHEMISTRY)**

How many moles of $\text{K}_2\text{Cr}_2\text{O}_7$ can be reduced by 1 mole of Sn^{2+} ?

(1) $\frac{2}{3}$ (2) $\frac{1}{6}$ (3) $\frac{1}{3}$

(4) 1

- Q52** What happens when a mixture of acetylene and hydrogen is passed over heated Lindlar's catalyst?

(1) Ethylene and water are formed

(2) Ethane and water are formed

[Android App](#)[iOS App](#)[PW Website](#)

- (3) Only ethylene is formed
 (4) Acetylene and ethane are formed

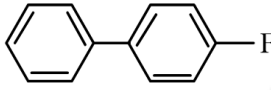
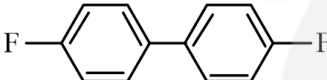
Q53 Time required to deposit one millimole of aluminium metal by the passage of 9.65 amperes through molten electrolyte containing aluminium ion is:

- (1) 30 s (2) 10 s
 (3) 30,000 s (4) 10,000 s

Q54 For $K_4[Fe(CN)_6]$, coordination number and oxidation state of central atom are respectively:

- (1) 6, +6 (2) 2, +2
 (3) 6, +2 (4) 2, +6

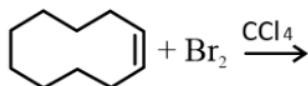
Q55 $Ph - NH_2 \xrightarrow[0^\circ C]{HNO_2} A \xrightarrow[BF_3]{HF} B \xrightarrow{\Delta} C$, C is :

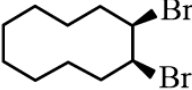
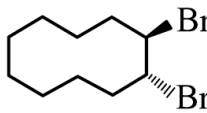
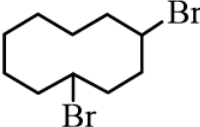
- (1) $Ph - \overset{+}{N} \equiv NBF_4^-$
 (2) 
 (3) 
 (4) $Ph - F$

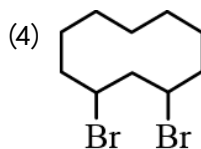
Q56 In the reaction,
 $VO + Fe_2O_3 \rightarrow FeO + V_2O_5$, the equivalent weight of V_2O_5 is equal to its:

- (1) M.Mass (2) M.Mass/8
 (3) M.Mass/6 (4) M.Mass/9

Q57



- (1) 
 (2) 
 (3) 



Q58 Electrochemical equivalent of an element is:

- (1) $\frac{\text{Atomic weight} \times \text{Valency}}{96500}$
 (2) $\frac{\text{Atomic weight} \times 96500}{\text{Valency}}$
 (3) $\frac{\text{Atomic weight}}{\text{Valency} \times 96500}$
 (4) $\frac{\text{Valency} \times 96500}{\text{Atomic weight}}$

Q59 Which of the following is diamagnetic?

- (1) $[Cu(NH_3)_4]^{2+}$
 (2) $[NiCl_4]^{2-}$
 (3) $[PtCl_4]^{2-}$
 (4) $[Cu(H_2O)_4]^{2+}$

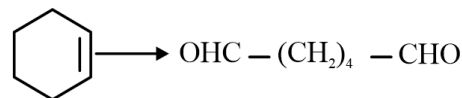
Q60 Standard reduction potentials of three metals X, Y and Z are -1.2 V, $+5$ V and -3.0 V respectively. The reducing power of these metals will be:

- (1) $Y > Z > X$
 (2) $Y > X > Z$
 (3) $Z > X > Y$
 (4) $X > Y > Z$

Q61 In acidic medium, H_2O_2 changes $Cr_2O_4^{2-}$ to CrO_5 which has two $(-O-O-)$ bonds. Oxidation state of Cr in $Cr_2O_7^{2-}$ and CrO_5 are respectively:

- (1) +6 and +5
 (2) +4 and +3
 (3) +6 and +6
 (4) +8 and -10

Q62 Select the reagent for the following reaction.



- (1) SeO_2
 (2) $O_3, Zn/H_2O$
 (3) $O_3, H_2O_2-CH_3COOH$
 (4) PCC

Q63 Among the following cells:

Leclanche cell (I); Nickel-Cadmium cell (II); Lead storage battery (III); Mercury cell (IV) primary cell are



- (1) I and II only
 (2) I and III only
 (3) II and III only
 (4) I and IV only

Q64 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion: Solvents play an important role in deciding the nucleophilicity.

Reason: Small ions are solvated best in aprotic solvents.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

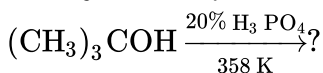
Q65 The structure formula of EDTA ion is:

- (1) $(^-\text{OOCCH}_2)_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{N}(\text{CH}_2\text{COO}^-)_2$
 (2) $\text{H}_2\text{NCH}_2 - \text{COO}^-$
 (3) $(^-\text{OOCCH}_2)_2\text{N} - \text{CH}_2 - \text{CH}_2 - \text{N}(\text{CH}_2\text{COO}^-)_2$
 (4) $\text{H}_2\ddot{\text{N}}(\text{CH}_2)_2\text{H}\ddot{\text{N}}\text{CH}_2 - \text{CH}_2\ddot{\text{N}}\text{H}(\text{CH}_2)_2\ddot{\text{N}}\text{H}_2$

Q66 In Br_3O_8 compound, overall oxidation number of bromine is:

- (1) 16/13 (2) 18/3
 (3) 24/3 (4) 16/3

Q67 Identify the main product in the reaction

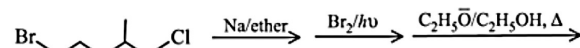


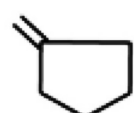
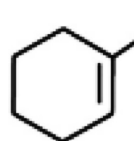
- (1) $(\text{CH}_3)_2\text{C} = \text{CH}_2$ (2) $(\text{CH}_3)_2\text{CHOH}$
 (3) $\text{CH}_3\text{CH} = \text{CHCH}_3$ (4) $\text{CH}_3\text{CH}_2\text{CH} = \text{CH}_2$

Q68 Which of the following oxides is used in lead storage battery?

- (1) PbO (2) PbO_2
 (3) Pb_2O_3 (4) Pb_3O_4

Q69 The major product of the following reaction is



- (1) 
 (2) 
 (3) 
 (4) 

Q70 A metal complex having composition $\text{Cr}(\text{NH}_3)_4\text{Cl}_2\text{Br}$ has been isolated in two forms X and Y. The form X reacts with AgNO_3 , to give a white precipitate readily soluble in aqueous ammonia, whereas Y gives a pale yellow precipitate partial soluble in aqueous ammonia X and Y are respectively:

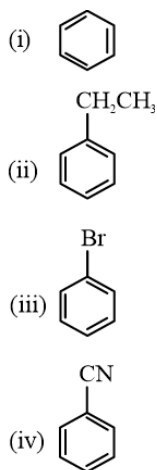
- (1) $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$, and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$
 (2) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$, and $[\text{Cr}(\text{NH}_3)_4\text{BrCl}]\text{Cl}$
 (3) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_2\text{Br}]\text{NH}_3$, and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$
 (4) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_2\text{Br}]\text{NH}_3$, and $[\text{Cr}(\text{NH}_3)_4\text{ClBr}]\text{Cl}$

Q71 In which of the following compounds oxidation state of chlorine has two different values?

- (1) CaCl_2 (2) NaCl
 (3) CaOCl_2 (4) CCl_4

Q72 Identify the **correct** order of reactivity in electrophilic substitution reaction of the following compounds:



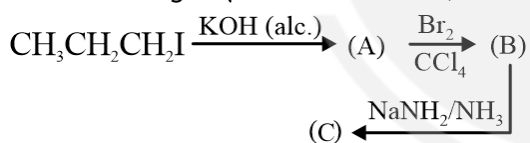


- (1) (i) > (ii) > (iii) > (iv)
 (2) (iv) > (iii) > (ii) > (i)
 (3) (ii) > (i) > (iii) > (iv)
 (4) (ii) > (iii) > (i) > (iv)

Q73 Which is **incorrect** about fuel cells?

- (1) Cells continuously run as long as fuels are supplied.
 (2) These are more efficient and free from pollution.
 (3) These are used to provide power and drinking water to astronauts in space programme.
 (4) Fuel cell has low efficiency.

Q74 In the following sequence of reactions,



The end product (C) is

- (1) Propene (2) Propanol
 (3) Propyne (4) Propyl amine

Q75 Among the following which are ambidentate ligands?

- A. NO_2^- B. NO_3^-
 C. EDTA^{4-} D. $\text{C}_2\text{O}_4^{2-}$
 E. SCN^- F. $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$
 (1) A and B (2) C and D
 (3) A and E (4) C and F

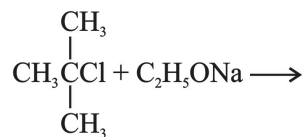
Q76 Arrange ICl , HI , I_2 and HIO_4 in decreasing order of oxidation state of iodine.

- (1) $\text{HIO}_4 > \text{ICl} > \text{HI} > \text{I}_2$
 (2) $\text{ICl} > \text{HIO}_4 > \text{I}_2 > \text{HI}$

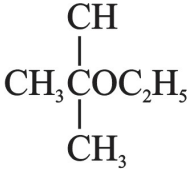
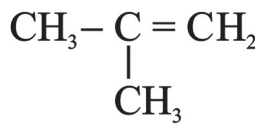
(3) $\text{HIO}_4 > \text{ICl} > \text{I}_2 > \text{HI}$

(4) $\text{ICl} > \text{HIO}_4 > \text{HI} > \text{I}_2$

Q77



Major product of this reaction is

- (1) 
 (2) 
 (3) 
 (4) None of these

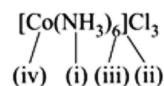
Q78 Mark the **correct** decreasing order of stability.

- (1) Aromatic > non-aromatic > anti-aromatic
 (2) Aromatic > anti-aromatic > non-aromatic
 (3) Non-aromatic > anti-aromatic > aromatic
 (4) Anti-aromatic > non-aromatic > aromatic

Q79 How many coulombs of electricity are required for the reduction of 1 mol of MnO_4^- to Mn^{2+} ?

- (1) $3.53 \times 10^5 \text{ C}$
 (2) $1.93 \times 10^5 \text{ C}$
 (3) $4.83 \times 10^5 \text{ C}$
 (4) $5.62 \times 10^5 \text{ C}$

Q80 Mark the **correct** labelling of different terms used in coordination compounds.



- (1) (i) Central atom, (ii) Ionisation sphere, (iii) Coordination number, (iv) Ligands
 (2) (i) Ligands, (ii) Coordination number, (iii) Valency, (iv) Ionisation sphere
 (3) (i) Ionisation sphere, (ii) Ligands, (iii) Coordination number, (iv) Central atom



- (4) (i) Ligands, (ii) Ionisation sphere, (iii) Coordination number, (iv) Central atom

Q81 Match List-I with List-II.

List-I (compound)		List-II (oxidation state of central atom)	
(A)	N_2O_5	(I)	-2
(B)	NaN_3	(II)	+5
(C)	NO	(III)	-1/3
(D)	N_2H_4	(IV)	+2

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
 (2) A-I, B-II, C-III, D-IV
 (3) A-IV, B-I, C-II, D-III
 (4) A-III, B-I, C-IV, D-II

Q82 Methane can be converted into ethane by the reaction:

- (1) chlorination followed by the reaction with alcoholic KOH.
 (2) chlorination followed by the reaction with aqueous KOH.
 (3) chlorination followed by Wurtz reaction.
 (4) chlorination followed by decarboxylation.

Q83 HA is a weak acid. At 25°C , the molar conductivity of 0.02 M HA is $150\ \Omega^{-1}\text{ cm}^2\text{ mol}^{-1}$. If its Λ_m° is $300\ \Omega^{-1}\text{ cm}^2\text{ mol}^{-1}$, the equilibrium constant of HA dissociation is:

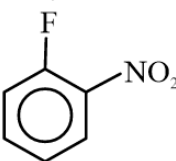
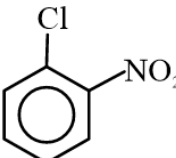
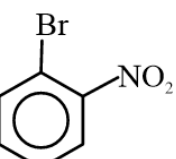
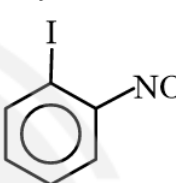
- (1) 0.001 M (2) 0.005 M
 (3) 0.01 M (4) 0.02 M

Q84 On reaction of 3-bromo-2, 3-dimethyl pentane with alc. KOH, three products as given below are formed. Arrange these in order of decreasing yield.

- (I) $(\text{CH}_3)_2\text{C}=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$
 (II) $\text{CH}_2=\text{C}(\text{C}_2\text{H}_5)\text{CH}(\text{CH}_3)_2$
 (III) $\text{CH}_3-\text{CH}=\text{C}(\text{CH}_3)\text{CH}(\text{CH}_3)_2$

- (1) I > II > III
 (2) I > III > II
 (3) III > II > I
 (4) all are formed in same percentage yield.

Q85 Which of the following compounds will undergo substitution of the halogen atom most readily with CH_3ONa in CH_3OH ?

- (1) 
 (2) 
 (3) 
 (4) 

Q86

SECTION - B (CHEMISTRY)

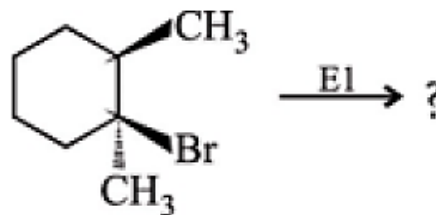
Which one of the following is an outer orbital complex and exhibits paramagnetic behaviour?

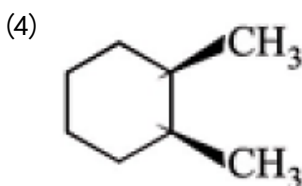
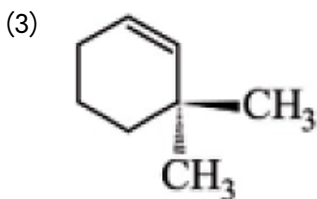
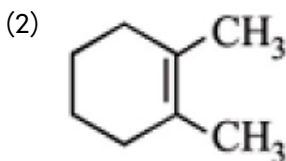
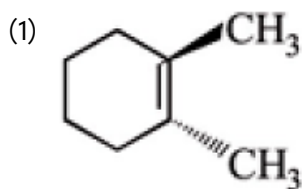
- (1) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ (2) $[\text{Zn}(\text{NH}_3)_6]^{2+}$
 (3) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (4) $[\text{Co}(\text{NH}_3)_6]^{3+}$

Q87 The specific conductance of a saturated solution of AgCl at 25°C is $1.821 \times 10^{-5}\text{ mho cm}^{-1}$. What is the solubility of AgCl in water, if limiting molar conductivity of AgCl is $130.26\text{ mho cm}^2\text{ mol}^{-1}$?

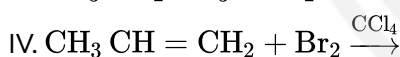
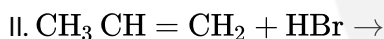
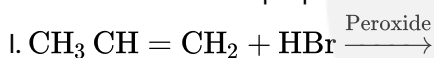
- (1) $1.89 \times 10^{-3}\text{ g L}^{-1}$
 (2) $2.78 \times 10^{-2}\text{ g L}^{-1}$
 (3) $2.004 \times 10^{-2}\text{ g L}^{-1}$
 (4) $1.43 \times 10^{-3}\text{ g L}^{-1}$

Q88 Choose the **correct** stereochemical presentation of the major product of the following reaction.





Q89 Select the reaction that would result in the formation of 2-bromopropane.



(1) I, II

(2) II, III

(3) I, III

(4) I, II, III

Q90 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cyclooctatetraene is a non-aromatic compound.

Reason R: Cyclooctatetraene follows $(4n + 2)\pi$ rule.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

Q91 Which among the following groups when attached to benzene ring will direct the incoming electrophile predominantly to the *meta* position?

A. $-\text{CH}_3$

B. $-\text{COOR}$

C. $-\text{NH}_3^+$

D. $-\text{CN}$

E. $-\text{NHCOCH}_3$

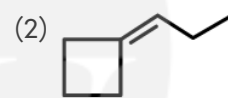
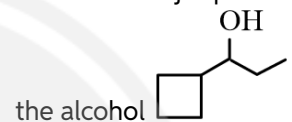
(1) C only

(2) A, B and E only

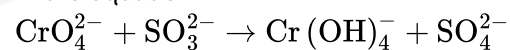
(3) B, C and D only

(4) C, D and E only

Q92 Select the major product of the dehydration of



Q93 In the equation



the oxidation number of Cr changes from:

(1) +6 to +2

(2) +6 to +3

(3) +8 to +4

(4) +4 to +3

Q94 The conductivity of $0.001028 \text{ mol L}^{-1}$ acetic acid is $4.95 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its dissociation constant, if Λ_m° for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$.

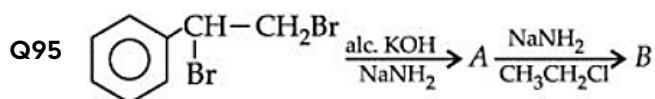
(1) $1.78 \times 10^{-5} \text{ mol L}^{-1}$

(2) $1.87 \times 10^{-5} \text{ mol L}^{-1}$

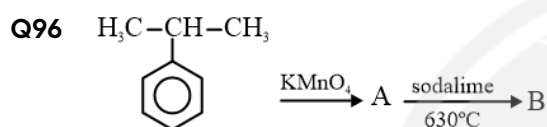
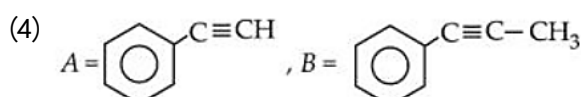
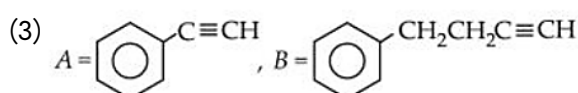
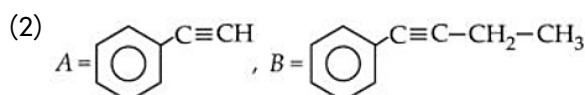
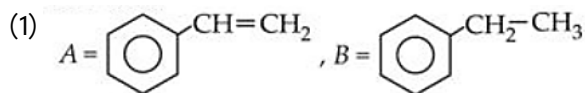
(3) $0.178 \times 10^{-5} \text{ mol L}^{-1}$

(4) $0.0178 \times 10^{-5} \text{ mol L}^{-1}$

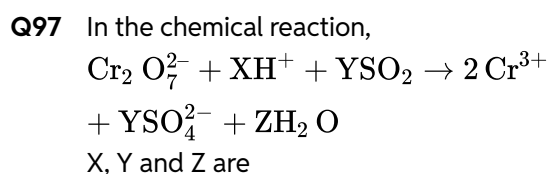
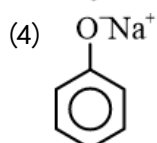
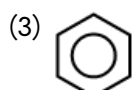
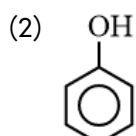
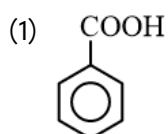




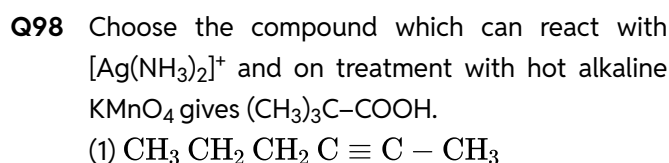
A and B are:



What is B?



- (1) 2, 3, 1 (2) 4, 1, 4
 (3) 3, 2, 3 (4) 2, 1, 2



- (2) $(\text{CH}_3)_2\text{CHCH}_2\text{C}\equiv\text{CH}$
 (3) $(\text{CH}_3)_3\text{C}-\text{C}\equiv\text{CH}$
 (4) $(\text{CH}_3)_3\text{C}-\text{C}\equiv\text{C}-\text{CH}_3$

Q99 Following limiting molar conductivities are given as;

$$\lambda_{\text{m}(\text{H}_2\text{SO}_4)}^\circ = x \text{ S cm}^2 \text{ mol}^{-1};$$

$$\lambda_{\text{m}(\text{K}_2\text{SO}_4)}^\circ = y \text{ S cm}^2 \text{ mol}^{-1};$$

$$\lambda_{\text{m}(\text{CH}_3\text{COOK})}^\circ = z \text{ S cm}^2 \text{ mol}^{-1};$$

λ_{m}° (in $\text{S cm}^2 \text{ mol}^{-1}$) for CH_3COOH will be:

- (1) $x - y + 2z$
 (2) $x + y - z$
 (3) $x - y + z$
 (4) $\frac{(x-y)}{2} + z$

Q100 Which of the following options are **correct** for $\text{Cs}[\text{FeCl}_4]$ complex?

- (I) Diamagnetic
 (II) Magnetic moment = 5.92 BM
 (III) Hybridisation- sp^3
 (IV) Paramagnetic
 (1) (I) and (II) only
 (2) (II), (III) and (IV) only
 (3) (II) and (IV) only
 (4) (I), (II) and (III) only

Q101

SECTION - A (BOTANY)

Which of the following has the same phyllotaxy as guava?

- (1) Sun flower (2) Mustard
 (3) *Calotropis* (4) *Alstonia*

Q102 Identify from the following which is/are related to epidermis?

- A. Trichome
 B. Root hair
 C. Guard cell
 D. Subsidiary cell

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D are correct
 (2) Only A, B and C are incorrect
 (3) Only A, B and C are correct
 (4) Only B and D are incorrect



Q103 The value of RQ will be less than one in which of the following substrates.

- (1) Tripalmitin and glucose
- (2) Glucose and fat
- (3) Protein and glucose
- (4) Tripalmitin and protein

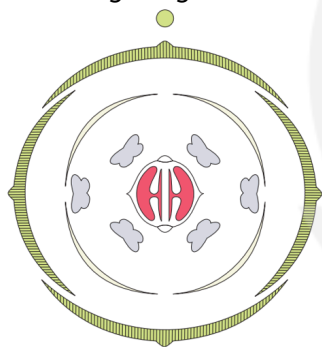
Q104 Select the **correct** statements about photosynthesis.

- A. Photosynthesis is a physico-chemical process.
- B. It is responsible for the release of oxygen.
- C. It is the primary source of all food on earth.
- D. Green plants are heterotrophs.

Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) B and D only
- (3) A and D only
- (4) A, B and C only

Q105 Select **correct** floral formula for the following floral diagram given below.



- (1) $\oplus \text{K}_{4+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$
- (2) $\oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(2)}$
- (3) $\oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+2} \underline{\text{G}}_{(2)}$
- (4) $\oplus \text{K}_{2+2} \text{C}_4 \text{A}_{2+4} \underline{\text{G}}_{(1)}$

Q106 Select the **mismatched** pair from the following for bulliform cell.

- (1) Turgid cell - Leaf surface is exposed
- (2) Flaccid cell - Leaf curl
- (3) Location - On abaxial surface
- (4) Size - Large

Q107 Select the **incorrect** statements.

- A. The fate of the pyruvate depends upon the availability of oxygen.
- B. The fate of the pyruvate depends upon the organism.
- C. The breaking of H-H bond in cellular respiration releases energy.
- D. The initial stage of cellular respiration takes place in the mitochondria.

Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) B and D only
- (3) C and D only
- (4) A, B and C only

Q108 Identify the **correct** statement related to growth.

- (1) Growth of pollen tube is measured in terms of surface area.
- (2) Growth in dorsiventral leaf is measured in terms of length.
- (3) Cells in watermelon increase by 3,50,000 times in an hour.
- (4) Cells of root apical meristem in maize can give rise to more than 17,500 new cells per hour.

Q109 Photosynthesis is essentially a light-dependent reaction was demonstrated in bacteria by:

- (1) Julius von Sachs.
- (2) T. W. Engelmann.
- (3) Cornelius van Neil.
- (4) Priestley.

Q110 Match **List-I** with **List-II**:

List-I		List-II	
(A)	Testa	(I)	Above the hilum
(B)	Tegmen	(II)	Outer layer of seed
(C)	Micropyle	(III)	Inner layer of seed
(D)	Aleurone	(IV)	Outer covering of endosperm

Choose the **correct** answer from the options given below:

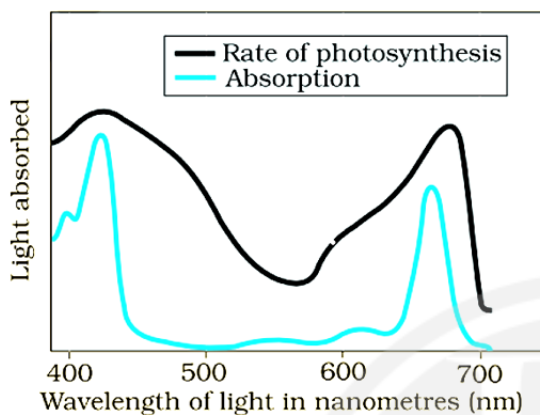
- (1) A-III, B-II, C-I, D-IV
- (2) A-II, B-III, C-I, D-IV
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-III, C-IV, D-I



Q111 Which of the following is **not** parenchymatous in the primary stems.

- (1) Cortex and pith
- (2) Pericycle and epidermis
- (3) Pith and medullary rays
- (4) Hypodermis

Q112 The given graph of rate of photosynthesis and absorption is **correct** for:



- (1) chlorophyll *b*
- (2) carotenoids
- (3) chlorophyll *a*
- (4) xanthophyll

Q113 Given below are two statements.

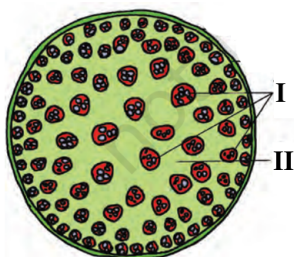
Statement I: The ground tissue system forms the main bulk of the plant.

Statement II: The vascular system is divided into three zones – cortex, pericycle and pith.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
- (2) Statement-I is incorrect and Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

Q114 In the following diagram of monocot stem, identify I and II respectively.



- (1) I-Ground tissue, II-Xylem
- (2) I-Vascular bundle, II-Ground tissue
- (3) I-Cambium, II-Ground tissue
- (4) I-Hypodermis, II-Ground tissue

Q115 Identify the **wrong** from the following.

- (1) Pedicel - Stalk of flower
- (2) Thalamus - Receptacle
- (3) Lamina - Leaf blade
- (4) Staminode - Fertile stamen

Q116 Given below are two statements:

Statement I: In arithmetic growth only one daughter cell continues to divide.

Statement II: In geometrical growth initial growth is fast and then there is lag phase.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q117 In dorsiventral leaf:

- (1) cuticle is absent in abaxial epidermis.
- (2) more stomata are present in adaxial surface.
- (3) mesophyll lacks chloroplast.
- (4) palisade parenchyma is made up of elongated cells.

Q118 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The rise in rate of respiration is called respiratory climactic.

Reason R: Gibberellins enhances the respiration rate during ripening of the fruits.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.



(4) Both A and R are true but R is not the correct explanation of A.

Q119 Cytochrome c is attached to the _____X_____ of the inner membrane of mitochondria and acts as a mobile carrier for transfer of electrons between complex _____Y_____.

Identify X and Y respectively.

- (1) X-inner surface: Y- III and IV
- (2) X-outer surface: Y- II and III
- (3) X-outer surface: Y- III and IV
- (4) X-inner surface: Y- IV and V

Q120 Given below are two statements.

Statement I: *Cassia* is zygomorphic with imbricate aestivation.

Statement II: Anterior petals in *Cassia* are keel.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct and Statement II is incorrect.
- (2) Statement I is incorrect and Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

Q121 Select **incorrect** for pericycle in stem.

- (1) Present on the inner side of the endodermis.
- (2) Present above the phloem.
- (3) Present in the form of semi-lunar patches.
- (4) Found in both monocot and dicot stem.

Q122 Read the following statements:

- A. Inhibitor-B is nothing but ethephon.
- B. Gibberellins cause fruits like apple to elongate and improve its shape.
- C. Gibberellins help to initiate rooting in stem cuttings.
- D. Auxins promote flowering in pineapples.

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D are correct
- (2) Only A, B and C are incorrect
- (3) Only A, B and C are correct
- (4) Only B and D are correct

Q123 Which of the following is natural auxin?

- (1) Indole-3-acetic acid
- (2) Naphthalene acetic acid
- (3) 2, 4-Dichlorophenoxy acetic acid
- (4) Both (1) and (2)

Q124 In Krebs cycle there are:

- (1) three points where NAD^+ is oxidised to $\text{NADH} + \text{H}^+$.
- (2) one point where FAD^+ is oxidised to FADH_2 .
- (3) three point where FAD^+ is reduced to FADH_2 .
- (4) three points where NAD^+ is reduced to $\text{NADH} + \text{H}^+$.

Q125 Identify the **correct** statement.

- (1) Pericarp is fruit wall.
- (2) Endocarp is outer fruit wall.
- (3) Epicarp is inner fruit wall.
- (4) Coconut develops from monocarpellary inferior ovary.

Q126 Select the vascular bundles in which cambium is observed .

- A. Open vascular bundles
- B. Radial vascular bundles in monocot root
- C. Closed vascular bundles
- D. All conjoint vascular bundles

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) A, C and D only
- (4) A only

Q127 Match **List-I** with **List-II**:

List-I		List-II (In C_4 plant)	
(A)	$\text{NADPH} + \text{H}^+$ formation	(I)	Calvin cycle
(B)	ATP formation only	(II)	OAA formation
(C)	RuBisCO is utilised	(III)	Stromal lamella
(D)	PEPcase is utilised	(IV)	Non cyclic photophosphorylation



Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
 (2) A-III, B-IV, C-II, D-I
 (3) A-II, B-IV, C-III, D-I
 (4) A-IV, B-III, C-I, D-II

Q128 Select the **correct** regarding respiration in plants.

- A. Presence of stomata
 B. Presence of lenticels
 C. Loose packing of parenchyma cells in leaves
 D. Each plant part takes care of its own gas-exchange needs.

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D
 (2) A, B and D only
 (3) A and C only
 (4) C only

Q129 Carboxylation is the most crucial step of the Calvin cycle where:

- (1) CO_2 is synthesised.
 (2) CO_2 is utilised.
 (3) ATP is utilised.
 (4) water is released as by product.

Q130 Rachis is present in:

- (1) silk cotton. (2) neem.
 (3) guava. (4) *Alstonia*.

Q131 Biosynthetic phase in C_3 plant release all **except**:

- (1) $\text{ADP} + \text{P}_i$
 (2) water.
 (3) hexose sugar.
 (4) oxaloacetic acid.

Q132 Plant growth regulator that respond to wounds and stresses of biotic and abiotic origin include:

- (1) auxin. (2) gibberellins.
 (3) cytokinin. (4) abscisic acid.

Q133 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The trichomes are present in the root system.

Reason R: The trichomes may be secretory.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q134 In yeast during fermentation all are produced **except**:

- (1) ethanol.
 (2) carbon dioxide.
 (3) water.
 (4) NAD^+ .

Q135 How many of the following are **correct**.

	Symmetry	Plant
A.	Actinomorphic	Pea
B.	Zygomorphic	Bean
C.	Bilateral	Mustard
D.	Radial	Gulmohar
E.	Asymmetrical	Chilli

Choose the most appropriate answer from the options given below:

- (1) Two (2) Four
 (3) Five (4) One

Q136

SECTION - B (BOTANY)

Polyarch xylem bundles are characteristic of:

- (1) monocot root and stem.
 (2) dicot stem and monocot root.
 (3) monocot stem and dicot root.
 (4) monocot root only.

Q137 In non-cyclic photophosphorylation splitting of the water molecule takes place:

- (1) near PSI.
 (2) on the inner side of the thylakoid membrane.
 (3) on the outer side of the chloroplast membrane.
 (4) on the outer side of the stromal lamella.

Q138 Match List-I with List-II:

List-I	List-II (Worked on)
--------	-------------------------

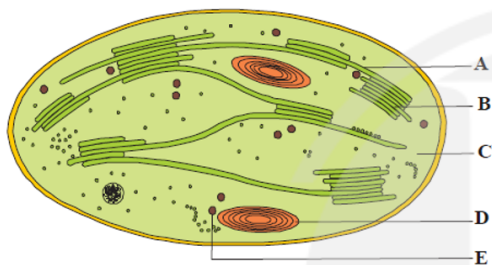


(A)	H.H. Cousins	(I)	Coleoptiles of oat seedlings.
(B)	F. Skoog	(II)	Rice seedlings
(C)	E. Kurosawa	(III)	Tobacco stem
(D)	F.W. Went	(IV)	Ripened oranges

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV
 (2) A-I, B-III, C-II, D-IV
 (3) A-IV, B-III, C-II, D-I
 (4) A-II, B-I, C-III, D-IV

Q139 The given figure shows the diagrammatic representation of a section of chloroplast. Few parts are marked as **A, B, C, D** and **E**.



Which parts are responsible for trapping the light energy and the synthesis of ATP.

Choose the most appropriate answer from the options given below:

- (1) A, B, C and D
 (2) A, B and D only
 (3) A and C only
 (4) A and B only

Q140 What is **true** about a monocot leaf?

- (1) Reticulate venation.
 (2) Absence of bulliform cells from epidermis.
 (3) Mesophyll not differentiated into palisade and spongy tissues.
 (4) Well differentiated mesophyll.

Q141 In the electron transport system, the final acceptor of proton is:

- (1) cytochrome *b*. (2) cytochrome *a₃*.
 (3) oxygen. (4) ubiquinone.

Q142 Which of the following is **not** a function of cytokinin?

- (1) Promotes apical dominance.
 (2) helps to produce chloroplast.

- (3) Promotes movement of nutrients.
 (4) Delay leaf senescence.

Q143 Identify the **mismatch** from the following.

- (1) Stele - Innermost layer of pericycle
 (2) Endodermis - Barrel shaped cells
 (3) Casparian strips - Endodermis
 (4) Initiation of lateral roots - Pericycle

Q144 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: In dorsiventral leaf spongy parenchyma is situated above the palisade cells.

Reason R: Spongy parenchyma cells are loosely arranged.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is not the correct explanation of A.

Q145 How many of the following are **correct** for flower.

- A. A flower is a modified shoot.
 B. When a shoot tip transforms into a flower, it always forms group of flowers.
 C. The arrangement of flowers on the floral axis is termed as inflorescence.
 D. Leaves and flower are borne together at apex of floral meristem.

Choose the most appropriate answer from the options given below:

- (1) One (2) Two
 (3) Three (4) Four

Q146 The given below leaf is observed in:



Adult

- (1) aquatic habitat in coriander.
 (2) aquatic habitat in larkspur.



- (3) aquatic habitat in cotton.
- (4) terrestrial habitat in buttercup.

Q147 Given below are two statements:

Statement I: ATP is synthesised during the conversion of phosphoenol pyruvate to pyruvic acid in glycolysis.

Statement II: Glycolysis is often referred to as the EMP pathway.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement-I is correct and Statement-II is incorrect.
- (2) Statement-I is incorrect and Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect

Q148 Select the **correct** features related to dicot root.

- A. Pith is large.
- B. Stele is present.
- C. A cambium ring develops between the xylem and phloem later in life.
- D. Conjunctive tissue is present.

Choose the most appropriate answer from the options given below:

- (1) A, B and C only
- (2) B, C and D only
- (3) A, B, C and D
- (4) B and C only

Q149 The formation of acetyl coenzyme A from one molecule of pyruvic acid produces (in addition to acetyl coenzyme A) one molecule of **A** one molecule of **B** and one molecule (ion) of **C**.

Identify **A**, **B** and **C** respectively.

- (1) A-H⁺, B-CO₂, C-NADH
- (2) A-CO₂, B-H⁺, C-NADH
- (3) A-NADH, B-O₂, C-H⁺
- (4) A-CO₂, B-NADH, C-H⁺

Q150 Identify the **correct** statement regarding root.

- (1) Root cap is present above region of maturation.

- (2) Cells of meristematic region are large with dense cytoplasm.
- (3) Cells of elongation help the root to grow in width.
- (4) Root cap is thimble like structure.

Q151

SECTION - A (ZOOLOGY)

Which of the following statements is **incorrect**?

- (1) Fibrinogens are needed for coagulation of blood.
- (2) Eosinophils are granulocytes that are involved in the allergic reactions.
- (3) The blood plasma along with the clotting factors is called serum.
- (4) Erythrocytes are the most abundant of all the cells in blood.

Q152 All of the following animals are uricotelic, **except**:

- (1) reptiles.
- (2) birds.
- (3) land snails.
- (4) aquatic insects.

Q153 Given below are two statements:

Statement I: Rh antigens of the foetus do not get exposed to the Rh^{-ve} blood of the mother in the first pregnancy as the two bloods are well separated by the placenta.

Statement II: A coagulam is formed mainly of a network of threads called fibrins in which dead and damaged formed elements of blood are trapped.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

Q154 Choose the **incorrect** statement.

- (1) The outer layer of the kidney is a tough capsule.



- (2) Two ureters and one urinary bladder are present in the human male.
- (3) Kidneys are placed close to the dorsal inner wall of the abdominal cavity.
- (4) Kidneys are situated between the levels of the third thoracic and last lumbar vertebra.

Q155 Which of the following is a powerful vasoconstrictor that increases the glomerular blood pressure and glomerular filtration rate?

- (1) Renin
- (2) Atrial natriuretic factor
- (3) Angiotensin II
- (4) All of these

Q156 Increase in sodium reabsorption in the distal convoluted tubule is due to:

- (1) increase in aldosterone levels.
- (2) increase in atrial natriuretic factor levels.
- (3) decrease in aldosterone levels.
- (4) decrease in antidiuretic hormone levels.

Q157 All of the following parts of the nephron allow the reabsorption of water through it, **except**:

- (1) proximal convoluted tubule.
- (2) distal convoluted tubule.
- (3) ascending limb of loop of Henle.
- (4) descending limb of loop of Henle.

Q158 Counter current mechanism helps in concentrating the urine in animals and it mainly operates on:

- A. Proximal convoluted tubule.
- B. Distal convoluted tubule.
- C. Loop of Henle.
- D. *Vasa recta*.

- (1) A, B and C only
- (2) C and D only
- (3) A and C only
- (4) B and D only

Q159 Which of the following features is **common** in both fish and crocodiles?

- (1) Presence of pulmonary circulation.
- (2) Numbers of chambers of heart.
- (3) Heart pumps only deoxygenated blood.
- (4) Presence of closed circulatory system.

Q160 Read the following statements(**A-D**).

- A. Human heart is myogenic.
- B. A special neural centre in the medulla oblongata of forebrain can moderate the cardiac function through the central nervous system.
- C. Heart failure is sometimes called congestive heart failure because congestion of the lungs is one of the main symptoms of this disease.
- D. Parasympathetic neural signals can increase the cardiac output.

Which of the following statements are **correct**?

- (1) A and B only
- (2) B and C only
- (3) C and D only
- (4) A and C only

Q161 Which of the following statements is **incorrect**?

- (1) Each kidney has nearly two million complex tubular structures called nephrons.
- (2) Nephrons are the functional units of the kidney.
- (3) Atrial systole increases the flow of blood into the ventricles by about 30 per cent.
- (4) Erythrocytes contain red colored iron containing pigment called haemoglobin.

Q162 Choose the set of **incorrect** statements.

- A. Bipolar neurons are found in the retina of eyes.
- B. Unipolar neurons have only one axon and one dendrite.
- C. Multipolar neurons are found only in the embryonic stages.
- D. Multipolar neurons have a cell body with one axon and two or more dendrites.

- (1) A and B only
- (2) B and C only
- (3) C and D only
- (4) A and D only

Q163 Read the following statements(**A-D**).

- A. The second heart sound is the dub sound which is heard due to the closure of atrio-ventricular valves.
- B. Lymph is a colourless fluid containing specialised lymphocytes which provide immune responses to the body.
- C. Bundle of His transmits action potential through the entire ventricular musculature and causes ventricular systole.



D. Human heart is the mesodermally derived organ, situated in the abdominal cavity.

How many of the following statements is/are true?

- | | |
|-----------|----------|
| (1) Three | (2) Four |
| (3) Two | (4) One |

Q164 Choose the **incorrectly** matched pair of hormones and their source.

- A. Antidiuretic hormone : Adenohypophysis.
 B. Aldosterone : Adrenal cortex.
 C. ANF : Atria of heart.
 D. Renin : Liver.

- | | |
|------------------|------------------|
| (1) A and B only | (2) B and C only |
| (3) C and D only | (4) A and D only |

Q165 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Birds and mammals exhibit double circulation.

Reason R: Two separate circulatory pathways are present in the heart of birds and mammals.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is a correct explanation of A.
 (4) Both A and R are true but R is not a correct explanation of A.

Q166 Which of the following pairs are **correctly** matched?

- A. Thigh bone : Femur, Two in both legs
 B. Ankle bones : Tarsals, Seven in each leg
 C. Phalanges : Palm bones, Eight in each arm
 D. Wrist bones : Carpals, ten in each arm

- | | |
|------------------|------------------|
| (1) A and C only | (2) B and C only |
| (3) A and B only | (4) C and D only |

Q167 Which of the following bones constitute the human skull?

- (1) Cranial and vertebrae bones
 (2) Facial and vertebrae bones
 (3) Cranial and facial bones
 (4) Sternum and ribs

Q168 Which of the following is an auto-immune disorder that affects neuromuscular junction?

- (1) Myasthenia gravis
 (2) Muscular dystrophy
 (3) Tetany
 (4) Arthritis

Q169 Identify the **incorrect** option w.r.t. number of bones of the vertebral column in an adult human beings.

- (1) Seven cervical vertebrae
 (2) Twelve thoracic vertebrae
 (3) Six lumbar vertebrae
 (4) Single fused sacral vertebra

Q170 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Haemodialysis helps in the removal of urea from the blood of human beings.

Reason R: Haemodialysis is the ultimate method in the correction of acute renal failures.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is a correct explanation of A.
 (4) Both A and R are true but R is not a correct explanation of A.

Q171 Which of the following statements is **incorrect**?

- (1) At the point of fusion of the coxal bones is a cavity called acetabulum to which the thigh bone articulates.
 (2) Ribs are connected ventrally to the vertebral column and dorsally to the sternum.
 (3) Fascicles are held together by a common collagenous connective tissue layer called fascia.
 (4) Flagellar movement helps in the maintenance of water current in the canal system of sponges.

Q172 Which of the following joints are present in the bones of cranium?

- (1) Fibrous joint



- (2) Ball and socket joint
 (3) Cartilaginous joint
 (4) Saddle joint

Q173 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Renal calculi	(I)	Genetic disorder
(B)	Glomerulonephritis	(II)	Inflammation of joints
(C)	Arthritis	(III)	Inflammation of glomeruli of kidney
(D)	Muscular dystrophy	(IV)	Stones formed within the kidney

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
 (2) A-II, B-III, C-I, D-IV
 (3) A-I, B-III, C-II, D-IV
 (4) A-IV, B-III, C-II, D-I

Q174 Given below are two statements:

Statement I: In axial skeleton, scapula is situated in the dorsal part of the thorax between the second and the seventh ribs.

Statement II: The periphery part of thick filament, not overlapped by thin filaments is called the 'H' zone.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Statement I and Statement II both are incorrect.

Q175 In human beings, ammonia produced by metabolism is converted into **A** in the liver and then excreted out by **B**.

Choose the option which fills the blank **correctly**.

A		B	
(1)	urea		liver

(2)	urea	kidneys
(3)	uric acid	liver
(4)	uric acid	kidneys

Q176 Which of the following statements is **incorrect**?

- (1) Blood is filtered by the kidneys per minute which constitute roughly $1/5^{\text{th}}$ of the blood pumped out by each ventricle of the heart in a minute.
 (2) Nearly 99 per cent of the glomerular filtrate has to be reabsorbed by the renal tubules.
 (3) Glucose and amino acids are reabsorbed by passive transport.
 (4) The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries.

Q177 The osmotic concentration of glomerular filtrate in the cortex region of kidneys is about ____ mOsmL⁻¹.

Choose the option which fills the blank **correctly**.

- (1) 900 (2) 600
 (3) 300 (4) 1200

Q178 Given below are two statements:

Statement I: Terrestrial adaptation necessitated the production of urea and uric acid for conservation of water.

Statement II: Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by the proximal convoluted tubule.

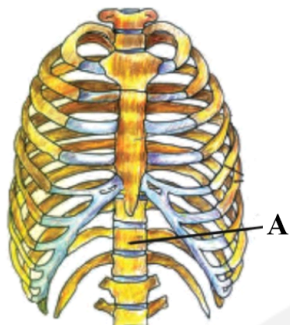
In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Statement I and Statement II both are incorrect.



- Q179** Which of the following options **correctly** describes the composition of a sarcomere?
- (1) One complete A band and a half I band
 - (2) Half A band and one complete I band
 - (3) One complete A band and two half I bands
 - (4) Two half A bands and two half I bands

- Q180** Identify the labeling “A” in the given below diagram and choose the **correct** option.



- (1) Rib has two articulation surfaces on its dorsal end and is hence called bicephalic.
- (2) Sternum is a flat bone on the ventral midline of thorax.
- (3) Vertebral column extends from the base of the skull and constitutes the main framework of the trunk.
- (4) Pelvic girdle consists of two coxal bones.

- Q181** Match **List-I** with **List-II**.

List-I		List-II	
(A)	Cerebrum	(I)	Controls the pituitary gland
(B)	Cerebellum	(II)	Controls vision and hearing
(C)	Hypothalamus	(III)	Seat of intelligence
(D)	Midbrain	(IV)	Maintains body posture

Choose the **correct** answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-III, B-I, C-II, D-IV
- (3) A-III, B-IV, C-I, D-II
- (4) A-III, B-I, C-IV, D-II

- Q182** A person entering an empty room suddenly finds a snake right in front on opening the door. Which of the following is likely to happen in his neuro-hormonal control system?

- (1) Hypothalamus activates the parasympathetic division of brain.
- (2) Sympathetic nervous system is activated releasing epinephrine and norepinephrine from adrenal cortex.
- (3) Sympathetic nervous system is activated releasing epinephrine and norepinephrine from adrenal medulla.
- (4) Neurotransmitters diffuse rapidly across the cleft and transmit a nerve impulse.

- Q183** A pregnant female delivers a baby, who suffers from stunted growth, mental retardation, low intelligence quotient and abnormal skin. This is the result of:

- (1) deficiency of iodine in diet.
- (2) low secretion of growth hormone.
- (3) cancer of the thyroid glands.
- (4) over secretion of pars distalis.

- Q184** Read the following statements (A-E).

- A. Along with the hypothalamus, limbic system is involved in the regulation of sexual behaviour.
- B. Binding of neurotransmitters to the receptor opens ion channels allowing the entry of ions which can generate a new potential in the post-synaptic neuron.
- C. Piamater is the innermost layer of meninges.
- D. The human brain is well protected by the cartilaginous skull.
- E. A deep cleft divides the cerebrum longitudinally into three halves.
- How many of the above statements is/are **correct**?

- (1) Two
- (2) One
- (3) Four
- (4) Three

- Q185** Which of the following is **not** a function of thyroid hormones?

- (1) Regulation of the basal metabolic rate.
- (2) Supporting the process of red blood cell formation.



- (3) Maintenance of water and electrolyte balance in the body.
 (4) Regulation of diurnal cycle.

Q186

SECTION - B (ZOOLOGY)

Which of the following features are **true** for red muscle fibres?

- A. They have more mitochondria.
 B. They contain large amounts of myoglobin.
 C. They have more sarcoplasmic reticulum.
 D. Mitochondria present in red muscle fibres can help in ATP production.

- (1) A, B and C only
 (2) B, C and D only
 (3) A, B and D only
 (4) C and D only

Q187 Given below are two statements:

Statement I: In polarised axonal membrane, the outer surface possesses a positive charge while its inner surface becomes negatively charged.

Statement II: Sodium-potassium pump actively transports 3Na^+ outwards for 2K^+ into the cell in the resting axonal membrane.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Statement I and Statement II both are incorrect.

Q188 Which of the following options is **correct** about corpus callosum?

- (1) It connects our forebrain and midbrain.
 (2) It is a band of nerves from midbrain to medulla oblongata.
 (3) It is a tract of nerve fibres which connects two cerebral hemispheres.
 (4) It connects thalamus and hypothalamus.

Q189 Read the following statement (A-D).

- A. In humans, four parathyroid glands are present on the back side of the thyroid gland, one pair

each in the two lobes of the thyroid gland.

B. The parathyroid glands secrete a peptide hormone called parathyroid hormone.

C. The thymus gland secretes the peptide hormones called thymosins.

D. The centrally located tissue in adrenal gland is called the adrenal cortex and outside this lies the adrenal medulla.

How many of the following statements is/are **incorrect**?

- (1) One (2) Two
 (3) Three (4) Four

Q190 Given below are two statements:

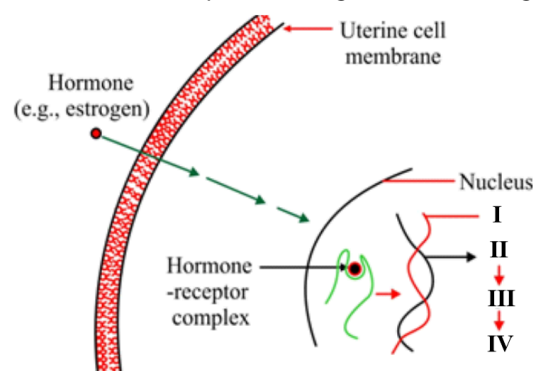
Statement I: Pancreas is a composite gland which acts as both exocrine and endocrine gland.

Statement II: In Addison's disease, overproduction of hormones by the adrenal cortex alters carbohydrate metabolism causing acute weakness and fatigue.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Statement I and Statement II both are incorrect.

Q191 The following diagram illustrates the representation of the mechanism of steroid hormone action and choose the **correct** option w.r.t the labelled part in the given below diagram.



- A. Proteins
 B. mRNA



- C. Genome
D. Physiological responses
(1) I→A, II→B, III→C, IV→D
(2) I→C, II→B, III→A, IV→D
(3) I→B, II→D, III→A, IV→C
(4) I→A, II→C, III→B, IV→D

Q192 Identify the circulatory disorder mainly on the basis of following characteristics and choose the **correct** option.

- A. Not enough oxygen reaches the heart muscles.
B. Can occur in men and women of any age.
C. Acute chest pain occurs in this disease.
(1) Atherosclerosis
(2) Angina pectoris
(3) Hypertension
(4) Cardiac arrest

Q193 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Ovary is the primary female sex organ that produces one ovum during each menstrual cycle.

Reason R: In human female ovary, estrogen is synthesised and secreted mainly by the growing ovarian follicles.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q194 Read the following statement (A-D).

- A. The PNS includes the brain and the spinal cord.
B. The PNS comprises of all the nerves of the body associated with the brain and spinal cord.
C. The nerve fibres of the PNS are of two types i.e., afferent fibres and efferent fibres.
D. The PNS is divided into two divisions called somatic neural system and autonomic neural system.

How many of the following features belong to the peripheral nervous system (PNS)?

- (1) Two (2) Three
(3) Four (4) One

Q195 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pineal gland	(I)	Thyroxine
(B)	Thyroid gland	(II)	Estrogen
(C)	Ovary	(III)	Epinephrine
(D)	Adrenal medulla	(IV)	Melatonin

Choose the **correct** answer from the options given below:

- (1) A-I, B-IV, C-II, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

Q196 Platelets, also called **A** are **B** produced from megakaryocytes (special cells in the bone marrow).

Choose the option which fills the blank **correctly**.

	A	B
(1)	thrombocytes	cell fragments
(2)	leucocytes	cell fragments
(3)	granulocytes	nucleated cells
(4)	monocytes	nucleated cells

Q197 Which of the following statements is **incorrect**?

- (1) In most of the invertebrates, simple tubular forms are the excretory structures.
(2) Some amount of urea may be retained in the kidney matrix of some of these animals to maintain a desired osmolarity.
(3) Nephridia plays a role in maintaining ionic fluid balances in annelids.
(4) Green glands perform the excretory function in cockroaches.



Q198 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The false ribs are also called floating ribs.

Reason R: The 11th and 12th pairs of ribs are not directly connected ventrally to the sternum and are called floating ribs.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is a correct explanation of A.
- (4) Both A and R are true but R is not a correct explanation of A.

Q199 Read the following statements (A-E).

- A. Plasma is a straw-coloured, viscous fluid constituting nearly 70% of the blood.
- B. Fats are absorbed only through blood in the lacteals present in the intestinal villi.
- C. In cardiac arrest, heart stops beating.
- D. Globulins primarily are involved in defense mechanisms of the body.

E. Mitral valve is present between left auricle and left ventricle.

Which of the above statements are **incorrect**?

- (1) A and B only
- (2) A, B and C only
- (3) C and E only
- (4) B, C, D and E only

Q200 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Ascending limb of loop of Henle allows transport of electrolyte actively and passively.

Reason R: As the concentrated filtrate pass upward, it gets diluted due to the passage of electrolytes to the medullary fluid.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.



SPACE FOR ROUGH WORK



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>